

CURRICULUM VITAE

Personal Profile

Name: **Kyo Cheol, Kang**

Nationality: Republic of Korea

Date of Birth: 30. 04. 2001

Status: M.S./Ph.D. integrated Student

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Education

Korea University, Seoul, South Korea

Mar. 2024 ~ Present

M.S./Ph.D. integrated in Electrical Engineering (GPA: 4.35 / 4.5), Expected Feb 2027

(Advisor: Prof. Byeong-Kwon Ju)

Konkuk University, Seoul, South Korea

Mar. 2020 ~ Feb. 2024

B.S. in Physics (GPA: 3.93 / 4.5)

Research Interests

- Organic Light-Emitting Diodes (OLEDs), Light Extraction and Photonic Structure Engineering, Microcavity and Multilayer Thin-Film Optics, Nanostructured Surfaces for Optoelectronic Devices, Computational Electromagnetic Modeling.

Research Experiences

- Graduate research assistant Mar. 2024 ~ Present
Display and nanosensor laboratory (Prof. Byeong-Kwon Ju), Korea University, Korea.
 - Achieved +32.3% EQE and +23.8% EL enhancement in blue TADF TEOLEDs through TiO₂ NP-embedded nanolaminated TFE (Samsung Display academia-industry collaboration); realized WVTR < 10⁻⁵ g m⁻² day⁻¹ with improved angular color stability via Mie-scattering-assisted light extraction and FDTD-validated optical design.
 - Research achieved +23.7 % Keff and 2× magnetic volume via DC/RF sputtering-based MTJ fabrication and high-T VSM/FMR analysis (Samsung Electronics collaboration); established reliability framework for automotive eMRAM.
 - Developed O₂ plasma-engineered wavy PDMS substrates for the Korea Institute for Advancement of Industry-Transition Type Inorganic Light-Emitting Display Specialist Training Program, achieving 97% EQE enhancement in green bottom-emitting OLEDs with >90% transmittance and tunable haze ; minimized angular color shift via FDTD-optimized light-extraction design.
 - Developed single-chamber CVD-based multilayered parylene films with tunable haze and enhanced substrate adhesion via pyrolysis temperature modulation; achieved up to 36% EL enhancement in OLEDs by suppressing substrate modes and promoting air-mode outcoupling through FDTD/TMM-validated scattering design while maintaining >82% total transmittance.
- Research Intern Jan. 2024 ~ Feb. 2024
Display and nanosensor laboratory (Prof. Byeong-Kwon Ju), Korea University, Korea.
 - Conducted preliminary fabrication and optoelectronic characterization of OLED devices, establishing workflows for J-V-L, EQE, and angular emission analysis to evaluate device efficiency, optical outcoupling, and operational stability.

Publication

1. K.C. Kang†, J.C. Bi†, J.Y. Park, H.S. Hwang, H. Lim, J. Kil, J.H. Kwon, B.K. Ju*, "Enhancing Optical Efficiency and Viewing Angle Characteristics of Thermally Activated Delayed Fluorescence Organic Light-Emitting Diodes via inverted TiO₂ Nanoparticle-Embedded PDMS Substrates" *Advanced Electronic Materials*, Feb. 2026 (*Revision*)
2. J.Y. Park†, K.C. Kang†, J.S. Lee†, J.C. Bi, H.S. Hwang, J. Kil, J.H. Kwon, H. Lim, B.K. Ju*, "Improving Efficiency and Color Purity of Blue Top-Emitting OLEDs: Distributed Bragg Reflectors for Surface Plasmon Polariton Suppression and Cavity-Resonance Enhancement", *ACS Applied Materials & Interfaces*, 18, 25, 35731–35741 (2026) IF: 7.8
3. J. Song†, J.C. Bi†, K.C. Kang, J. Kil, J.H. Kwon, S. Hwang, J.S. Lee, H. Lim, B.K. Ju*, "Enhancing color purity and efficiency of blue TADF OLEDs with MoO₃/Al₂O₃-based distributed Bragg reflector structures", *Optics Express*, 33(20), 42498–42511(2025) IF: 3.3
4. S.Y. Baek†, Y. Choi†, H.B. Kim, K.C. Kang, Y.S. Lee, A. Raji, M.G. Jung, S.Y. Boo, W. Jung, J. Lee, J.H. Lee*, and B.K. Ju*, "Single-Chamber Deposited Multilayered Parylene Films with Enhanced Visibility and Substrate Adhesion for Protective Device Coating", *Advanced Optical Materials*, e01820(2025) IF: 7.2
5. J.S. Lee, J.Y. Park, J.C. Bi, S. Park, J. Kil, K.C. Kang, S. Hwang, J.H. Kwon, J. Song, H. Lim, and B.K. Ju*, "Enhancing efficiency of blue thermally activated delayed fluorescence organic light-emitting device using a high refractive index material-based light extraction layer", *Journal of Luminescence*, 286 121399 (2025) IF: 3.8
6. D. G. Lim, J.Y. Park, Y. Ham, J. Kang, K.C. Kang, J.C. Bi, S. Park, J. Song, J.S. Lee, H. Lim, S.H. Shin, S.J. Kim, S.W. Han, and B.K. Ju, "2D Hole-Arrayed Double-Anode Structure Exciting Surface Plasmon Polaritons for Enhancing Outcoupling Efficiency of Organic Light-Emitting Diodes on Silicon Wafers (OLEDoS)", *ACS Omega*, 10(16), 16300–16308 (2025). IF: 5.2
7. J.C. Bi, J.Y. Park, S. Lee, S. Park, J. Song, J.S. Lee, K.C. Kang, K. Kim, D. Lim, Y.W. Park*, and B.K. Ju*, "TiO₂-Nanoparticle-Embedded Thin-Film Encapsulation for Blue Thermally Activated Delayed Fluorescence Top-Emitting Organic Light-Emitting Devices", *ACS Applied Materials & Interfaces* (2025). IF: 7.8
8. J.C. Bi†, K.C. Kang†, J.Y. Park, J. Song, J.S. Lee, H. Lim, Y.W. Park*, and B.K. Ju*, "Enhancement of Light Extraction Efficiency Using Wavy-Patterned PDMS Substrates", *Nanomaterials*, 15(3), 198 (2025). IF: 4.8

International Conferences

1. J.S. Lee, J.Y. Park, S. Park, K.C. Kang, and B.K. Ju, "Enhancing Efficiency of Blue Phosphorescent Organic Light Emitting Device Using a High Refractive Index Material-Based Light Extraction Layer", *The 22nd International Nanotech Symposium (NANO KOREA 2024)*, Goyang, Korea, TS11F_P_20 (2024.07.03–2024.07.05).
2. J.C. Bi, K.C. Kang, J.Y. Park, S. Park, J. Song, and B.K. Ju, "PDMS Substrate via Oxygen Plasma Treatment: Fabrication and Characterization", *The 22nd International Nanotech Symposium (NANO KOREA 2024)*, Goyang, Korea, TS11F_P_4 (2024.07.03–2024.07.05).
3. S. Park, J.Y. Park, J.C. Bi, J. Song, J.S. Lee, K.C. Kang, S.H. Shin, and B.K. Ju, "Shifting the Emission Peak of Organic Light-Emitting Diodes Utilizing Meta-Surface Reflection Phase Delay", *The 24th International Meeting on Information Display (IMID 2024)*, Jeju, Korea, 09_1058 (2024.08.20–2024.08.23).
4. J. Kil, J.Y. Park, J.S. Lee, K.C. Kang, S. Hwang, and B.K. Ju, "Application of a DBR Reflector for High-Efficiency Top-Emission OLEDs", *The 23rd International Nanotech Symposium (NANO KOREA 2025)*, Goyang, Korea, TS02F_P_30 (2025.07.02–2025.07.04).
5. J.H. Kwon, H. Lim, J.Y. Park, J.C. Bi, J. Song, K.C. Kang, and B.K. Ju, "Efficiency Improvement in Phosphorescent WOLEDs Employing a Multiple EML RGB Stacked Structure", *The 23rd International Nanotech Symposium (NANO KOREA 2025)*, Goyang, Korea, TS02F_P_29 (2025.07.02–2025.07.04).
6. S. Hwang, K.C. Kang, J.C. Bi, J. Song, "Enhancing Blue Thermally Activated Delayed Fluorescence OLEDs Efficiency with Corrugated PDMS: Design and Optical Analysis", *The International Conference on Electrical, Computer, Communications and Mechatronics Engineering (ICECCME 2025)*, Zanzibar, Tanzania, 779 (2025.10.16–2025.10.19)

Technical Skills

- Fabrication: DC/RF sputtering, thermal evaporation, ALD, CVD, RIE, ICP-RIE, mask aligner, spin-coating, O₂ plasma, wire-bonding, Laser Interference Lithography
- Characterization: 4-point probe, TEM, SEM, AFM, XPS, Spectroradiometer, High Voltage Source Measure Unit, UPS, Transient EL, Transient PL, Impedance Measurement.
- Data Analysis & Simulation: Python, C++, MATLAB, PLC, Ansys Lumerical FDTD, Image J,
- Java script, Abaqus, statistical modeling, process & reliability data analysis
- Integration & Reliability: High-efficiency TADF OLED operation validation, LT50 lifetime & bias/thermal stability testing, WVTR/Ca-test-based encapsulation reliability evaluation, angular color stability verification, and device/process optimization for microcavity & light-extraction control.

Personal Skills

- Hard working and committed.
- Keeping a positive attitude in all situations.
- Good communication skill.
- Quick grasping ability.
- Ability to work in a team as well coordinating the team.

Declaration

I declare that the foregoing information is correct and complete to the best of my knowledge and belief and nothing has been concealed or distorted.

K.C. Kang

Kyo-Cheol Kang

REFERENCES

- Prof. Byeong-Kwon Ju (M.S./Ph.D. integrated Program Supervisor)
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